DCCN Lab Assignment - 4

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1. **What is the IP address and TCP port number used by the client computer (source) that is transferring the file to gaia.cs.umass.edu?**
   The IP address and TCP port number used by the client computer (source) to transfer the file to *gaia.cs.umass.edu* are:
   IP Address: 192.168.1.102
   TCP Port Number: 1161

2. **What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?**
   The IP address of *gaia.cs.umass.edu* and the TCP port number on which it sends and receives TCP segments for this connection are:
   IP Address : 128.119.245.12
   Port Number : 80

3. **What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?**
   The IP address and TCP port number used by the client computer (source) to transfer the file to *gaia.cs.umass.edu* are:
   IP Address : 192.168.1.102
   Port Number : 1161

4. **What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? What is it in the segment that identifies the segment as a SYN segment?**
   The sequence number of the TCP SYN segment used to initiate the TCP connection is:
   Sequence Number: 0   (relative sequence number)
   Sequence Number (raw): 232129012

The segment is identified as a SYN segment by the SYN flag being set in the TCP flags field (SYN = 1, ACK = 0)

5. **What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value? What is it in the segment that identifies the segment as a SYNACK segment?**
   The sequence number of the SYN-ACK segment sent by *gaia.cs.umass.edu* in response to the SYN segment is:
   - Relative Sequence Number: 0
   - Raw Sequence Number: 883061785

   The acknowledgement number in the SYN-ACK segment is:
   - Relative Acknowledgement Number: 1
   - Raw Acknowledgement Number: 232129013

   The acknowledgement number is determined by incrementing the sequence number of the received SYN segment by one, since the SYN consumes one sequence number.

   The segment is identified as a SYN-ACK segment by both the SYN and ACK flags being set in the TCP flags field.

6. **What is the sequence number of the TCP segment containing the HTTP POST command?**
   The sequence number of the TCP segment containing the HTTP POST command is:
   Sequence Number: 164041   (relative sequence number)
   Sequence Number (raw): 232293053

7. **Consider the TCP segment containing the HTTP POST as the first segment in the TCP connection. What is the length of each of the first six TCP segments?**
   Considering the TCP segment containing the HTTP POST command as the first segment in the connection, the lengths of the first six TCP segments are:
   Segment 1 (HTTP POST) = 50 bytes

Segment 2 (TCP ACK=162039) = 0 bytes
Segment 3 (TCP ACK=162041) = 0 bytes
Segment 4 (TCP ACK=162091) = 0 bytes
Segment 5 (HTTP OK) = 730 bytes
Segment 6 (TCP ACK=731) = 0 bytes

8. **What is the EstimatedRTT value after the receipt of each ACK? Assume that the value of the EstimatedRTT is equal to the measured RTT for the first segment.**
The Estimated Round Trip Time (EstimatedRTT) is calculated using the formula:
EstimatedRTT = (1−α) * EstimatedRTT + α * SampleRTT, where α=0.125

As specified, the EstimatedRTT is initialized to the first measured SampleRTT.The first SampleRTT observed is approximately 30 ms, so:
Initial EstimatedRTT = 30 ms

As additional acknowledgements are received, the EstimatedRTT evolves as follows:
After the second ACK: approximately 35 ms
After the third ACK: approximately 45.6 ms
After the fourth ACK: approximately 63.65 ms
After the fifth ACK: approximately 88.2 ms

With further ACKs, the EstimatedRTT converges to approximately 180–200 ms, effectively
smoothing the large variations observed in the SampleRTT values.